Lecture 2
Matrices Operations

2.1 Operations with Matrices

2.2 Properties of Matrix Operations

2.3 The Inverse of a Matrix

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2.1 Operations with Matrices

- Matrix:

$$A=[a_{ij}]=\begin{bmatrix} a_{11} & a_{12} & a_{13} & \cdots & a_{1n} \\ a_{21} & a_{22} & a_{23} & \cdots & a_{2n} \\ a_{31} & a_{32} & a_{33} & \cdots & a_{3n} \\ \vdots & \vdots & \vdots & & \vdots \\ a_{m1} & a_{m2} & a_{m3} & \cdots & a_{mn} \end{bmatrix}_{m\times n} \in M_{m\times n}$$

(i,j)-th entry: a_{ij}

Number of rows: m

Number of columns: n

Size: $m\times n$

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- *i*-th row vector

$$r_i = \begin{bmatrix} a_{i1} & a_{i2} & \cdots & a_{in} \end{bmatrix}$$ row matrix

- *j*-th column vector

$$c_j = \begin{bmatrix} a_{1j} \\ a_{2j} \\ \vdots \\ a_{mj} \end{bmatrix}$$ column matrix

- Square matrix: $m = n$

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- Example: Partitioned matrices:

$$A = \begin{bmatrix} a_{11} & a_{12} & a_{13} & a_{14} \\ a_{21} & a_{22} & a_{23} & a_{24} \\ a_{31} & a_{32} & a_{33} & a_{34} \end{bmatrix} = \begin{bmatrix} r_1 \\ r_2 \\ r_3 \end{bmatrix}$$

$$A = \begin{bmatrix} a_{11} & a_{12} & a_{13} & a_{14} \\ a_{21} & a_{22} & a_{23} & a_{24} \\ a_{31} & a_{32} & a_{33} & a_{34} \end{bmatrix} = \begin{bmatrix} c_1 & c_2 & c_3 & c_4 \end{bmatrix}$$

$$A = \begin{bmatrix} a_{11} & a_{12} & a_{13} & a_{14} \\ a_{21} & a_{22} & a_{23} & a_{24} \\ a_{31} & a_{32} & a_{33} & a_{34} \end{bmatrix} = \begin{bmatrix} A_{11} & A_{12} \\ A_{21} & A_{22} \end{bmatrix}$$

submatrix

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- Diagonal matrix:

$$A = diag(d_1, d_2, \cdots, d_n) = \begin{bmatrix} d_1 & 0 & \cdots & 0 \\ 0 & d_2 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & d_n \end{bmatrix} \in M_{n\times n}$$

- Trace:

If $A = [a_{ij}]_{n\times n}$

Then $Tr(A) = a_{11} + a_{22} + \cdots + a_{nn}$

- Ex:

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix} = \begin{bmatrix} r_1 \\ r_2 \end{bmatrix}$$

$$\Rightarrow \quad r_1 = [1 \quad 2 \quad 3], \quad r_2 = [4 \quad 5 \quad 6]$$

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix} = [c_1 \quad c_2 \quad c_3]$$

$$\Rightarrow \quad c_1 = \begin{bmatrix} 1 \\ 4 \end{bmatrix}, \; c_2 = \begin{bmatrix} 2 \\ 5 \end{bmatrix}, \; c_3 = \begin{bmatrix} 3 \\ 6 \end{bmatrix}$$

- Equal matrix:

If $A=[a_{ij}]_{m\times n}$, $B=[b_{ij}]_{m\times n}$

Then $A=B$ if and only if $a_{ij}=b_{ij}\ \ \forall 1\le i\le m,\ 1\le j\le n$

- Ex 1: (Equal matrix)

$$A=\begin{bmatrix}1 & 2\\ 3 & 4\end{bmatrix} \qquad B=\begin{bmatrix}a & b\\ c & d\end{bmatrix}$$

If $A=B$

Then $a=1,\ b=2,\ c=3,\ d=4$

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- Matrix addition:

If $A=[a_{ij}]_{m\times n}$, $B=[b_{ij}]_{m\times n}$

Then $A+B=[a_{ij}]_{m\times n}+[b_{ij}]_{m\times n}=[a_{ij}+b_{ij}]_{m\times n}$

- Ex 2: (Matrix addition)

$$\begin{bmatrix}-1 & 2\\ 0 & 1\end{bmatrix}+\begin{bmatrix}1 & 3\\ -1 & 2\end{bmatrix}=\begin{bmatrix}-1+1 & 2+3\\ 0-1 & 1+2\end{bmatrix}=\begin{bmatrix}0 & 5\\ -1 & 3\end{bmatrix}$$

$$\begin{bmatrix}1\\ -3\\ -2\end{bmatrix}+\begin{bmatrix}-1\\ 3\\ 2\end{bmatrix}=\begin{bmatrix}1-1\\ -3+3\\ -2+2\end{bmatrix}=\begin{bmatrix}0\\ 0\\ 0\end{bmatrix}$$

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- Scalar multiplication:

  If $A=[a_{ij}]_{m\times n}$, c : scalar

  Then $cA=[ca_{ij}]_{m\times n}$

- Matrix subtraction:

  $A-B=A+(-1)B$

- Ex 3: (Scalar multiplication and matrix subtraction)

$$A=\begin{bmatrix}1&2&4\\-3&0&-1\\2&1&2\end{bmatrix} \qquad B=\begin{bmatrix}2&0&0\\1&-4&3\\-1&3&2\end{bmatrix}$$

Find $(a)\ 3A$, $(b)\ -B$, $(c)\ 3A-B$

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Sol:

(a)

$$3A=3\begin{bmatrix}1&2&4\\-3&0&-1\\2&1&2\end{bmatrix}=\begin{bmatrix}3(1)&3(2)&3(4)\\3(-3)&3(0)&3(-1)\\3(2)&3(1)&3(2)\end{bmatrix}=\begin{bmatrix}3&6&12\\-9&0&-3\\6&3&6\end{bmatrix}$$

(b)

$$-B=(-1)\begin{bmatrix}2&0&0\\1&-4&3\\-1&3&2\end{bmatrix}=\begin{bmatrix}-2&0&0\\-1&4&-3\\1&-3&-2\end{bmatrix}$$

(c)

$$3A-B=\begin{bmatrix}3&6&12\\-9&0&-3\\6&3&6\end{bmatrix}-\begin{bmatrix}2&0&0\\1&-4&3\\-1&3&2\end{bmatrix}=\begin{bmatrix}1&6&12\\-10&4&-6\\7&0&4\end{bmatrix}$$

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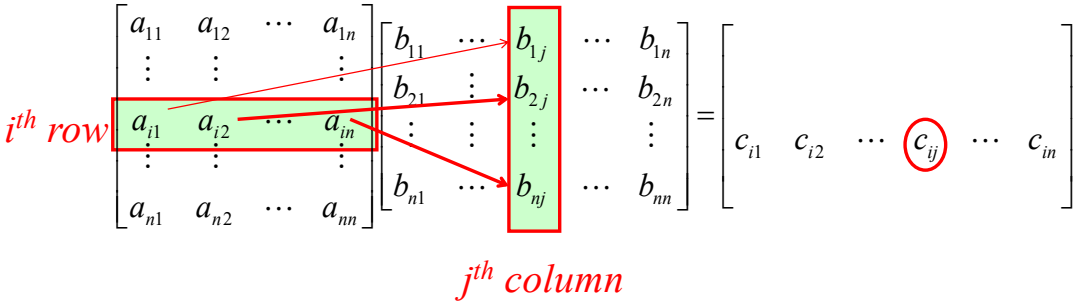
- Matrix multiplication:

If $A=[a_{ij}]_{m\times n}$, $B=[b_{ij}]_{n\times p}$

Then $AB=[a_{ij}]_{m\times n}[b_{ij}]_{n\times p}=[c_{ij}]_{m\times p}$

Size of AB

where $c_{ij}=\sum_{k=1}^{n}a_{ik}b_{kj}=a_{i1}b_{1j}+a_{i2}b_{2j}+\cdots+a_{in}b_{nj}$



- Notes: (1) $A+B=B+A$, (2) $AB\neq BA$

- Ex 4: (Find AB)

$$A=\begin{bmatrix}-1 & 3\\ 4 & -2\\ 5 & 0\end{bmatrix}\qquad B=\begin{bmatrix}-3 & 2\\ -4 & 1\end{bmatrix}$$

Sol:

$$AB=\begin{bmatrix}(-1)(-3)+(3)(-4) & (-1)(2)+(3)(1)\\ (4)(-3)+(-2)(-4) & (4)(2)+(-2)(1)\\ (5)(-3)+(0)(-4) & (5)(2)+(0)(1)\end{bmatrix}$$

$$=\begin{bmatrix}-9 & 1\\ -4 & 6\\ -15 & 10\end{bmatrix}$$

- Matrix form of a system of linear equations:

$$\begin{cases} a_{11}x_1 + a_{12}x_2 + \cdots + a_{1n}x_n = b_1 \\ a_{21}x_1 + a_{22}x_2 + \cdots + a_{2n}x_n = b_2 \\ \vdots \\ a_{m1}x_1 + a_{m2}x_2 + \cdots + a_{mn}x_n = b_m \end{cases}$$

m linear equations

$$\Downarrow$$

$$\begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \vdots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix} = \begin{bmatrix} b_1 \\ b_2 \\ \vdots \\ b_m \end{bmatrix}$$

$$\| \qquad \| \qquad \|$$

$$A \qquad x \qquad b$$

Single matrix equation

$$\underset{m\times n}{A}\ \underset{n\times 1}{x} = \underset{m\times 1}{b}$$

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2.2 Properties of Matrix Operations

- Three basic matrix operations:
  - (1) matrix addition
  - (2) scalar multiplication
  - (3) matrix multiplication

- Zero matrix:

$$0_{m\times n} = \begin{bmatrix} 0 & 0 & \cdots & 0 \\ 0 & 0 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & 0 \end{bmatrix} \in M_{m\times n}$$

- Identity matrix of order n:

$$I_n = \begin{bmatrix} 1 & 0 & \cdots & 0 \\ 0 & 1 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & 1 \end{bmatrix} \in M_{n\times n}$$

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- Properties of matrix addition and scalar multiplication:

If $A, B, C \in M_{m\times n}$, c, d : scalar

Then (1) $A+B = B+A$

(2) $A+(B+C)=(A+B)+C$

(3) $(cd)A = c(dA)$

(4) $1A = A$

(5) $c(A+B) = cA + cB$

(6) $(c+d)A = cA + dA$

- Properties of zero matrices:

If $A \in M_{m\times n}$, c : scalar

Then (1) $A + 0_{m\times n} = A$

(2) $A + (-A) = 0_{m\times n}$

(3) $cA = 0_{m\times n} \Rightarrow c = 0$ or $A = 0_{m\times n}$

- Notes:

(1) $0_{m\times n}$: **the additive identity** for the set of all $m\times n$ matrices — identity of the addition

(2) $-A$: **the additive inverse** of A — inverse of the addition

- Properties of identity matrices:

If $A \in M_{n\times n}$, I_n : identity matrix

Then $A.I_n = I_n.A = A.$

- Properties of matrix multiplication:

(1) $A.(B.C) = (A.B).C$ associativity

(2) $A.(B + C) = A.B + A.C$ distributivity

(3) $(A + B).C = A.C + B.C$ distributivity

(4) $c(A.B) = (cA).B = A.(cB)$

(5) $A.B \neq B.A$ non-commutativity

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- Transpose of a matrix:

$$\text{If } A = \begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \vdots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{bmatrix} \in M_{m\times n}$$

$$\text{Then } A^T = \begin{bmatrix} a_{11} & a_{21} & \cdots & a_{m1} \\ a_{12} & a_{22} & \cdots & a_{m2} \\ \vdots & \vdots & \vdots & \vdots \\ a_{1n} & a_{2n} & \cdots & a_{mn} \end{bmatrix} \in M_{n\times m}$$

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- Ex 8: (Find the transpose of the following matrix)

$$(a)\ A=\begin{bmatrix}2\\8\end{bmatrix} \qquad (b)\ A=\begin{bmatrix}1&2&3\\4&5&6\\7&8&9\end{bmatrix} \qquad (c)\ A=\begin{bmatrix}0&1\\2&4\\1&-1\end{bmatrix}$$

Sol: (a) $A=\begin{bmatrix}2\\8\end{bmatrix} \Rightarrow A^T=\begin{bmatrix}2&8\end{bmatrix}$

(b) $A=\begin{bmatrix}1&2&3\\4&5&6\\7&8&9\end{bmatrix} \Rightarrow A^T=\begin{bmatrix}1&4&7\\2&5&8\\3&6&9\end{bmatrix}$

(c) $A=\begin{bmatrix}0&1\\2&4\\1&-1\end{bmatrix} \Rightarrow A^T=\begin{bmatrix}0&2&1\\1&4&-1\end{bmatrix}$

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- Properties of transposes:

(1) $(A^T)^T=A$

(2) $(A+B)^T=A^T+B^T$

(3) $(cA)^T=c(A^T)$

(4) $(AB)^T=B^TA^T$

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- Symmetric matrix:

A square matrix A is **symmetric** if $A = A^T$

- Ex:

If $A = \begin{bmatrix} 1 & 2 & 3 \\ a & 4 & 5 \\ b & c & 6 \end{bmatrix}$ is symmetric, find a, b, c?

Sol:

$$A = \begin{bmatrix} 1 & 2 & 3 \\ a & 4 & 5 \\ b & c & 6 \end{bmatrix} \quad A^T = \begin{bmatrix} 1 & a & b \\ 2 & 4 & c \\ 3 & 5 & 6 \end{bmatrix}$$

$$A = A^T$$
$$\Rightarrow a = 2, b = 3, c = 5$$

- Note: AA^T is symmetric
- Note: A^TA is symmetric
- Matrix: in general

$$AB \neq BA$$

Let's consider the case when A is an $m \times n$ matrix and B is an $n \times p$ matrix.

We have 3 situations:

(1) If $m \neq p$, then AB is defined, BA is undefined.

(2) If $m = p, m \neq n$, then $AB \in M_{m \times m}$, $BA \in M_{n \times n}$ (Sizes are not the same)

(3) If $m = p = n$, then $AB \in M_{m \times m}$, $BA \in M_{m \times m}$

(Sizes are the same, but matrices are not equal)

- Ex 4:

Show that AB and BA are not equal for the matrices.

$$A=\begin{bmatrix}1 & 3\\ 2 & -1\end{bmatrix} \quad \text{and} \quad B=\begin{bmatrix}2 & -1\\ 0 & 2\end{bmatrix}$$

Sol:

$$AB=\begin{bmatrix}1 & 3\\ 2 & -1\end{bmatrix}\begin{bmatrix}2 & -1\\ 0 & 2\end{bmatrix}=\begin{bmatrix}2 & 5\\ 4 & -4\end{bmatrix}$$

$$BA=\begin{bmatrix}2 & -1\\ 0 & 2\end{bmatrix}\begin{bmatrix}1 & 3\\ 2 & -1\end{bmatrix}=\begin{bmatrix}0 & 7\\ 4 & -2\end{bmatrix}$$

- Conclusion: $AB \neq BA$

- For real numbers:

$ac = bc,\ c \neq 0$

$\Rightarrow\ a = b$ (Cancellation law)

- For matrices:

$AC = BC \quad C \neq 0$

In general, we can not imply that $A = B$

(1) If C is invertible, then $A = B$

(2) If C is not invertible, then $A \neq B$ (Cancellation is not valid)

- Ex 5: (An example in which cancellation is not valid)

Show that $AC=BC$

$$A=\begin{bmatrix}1 & 3\\ 0 & 1\end{bmatrix},\quad B=\begin{bmatrix}2 & 4\\ 2 & 3\end{bmatrix},\quad C=\begin{bmatrix}1 & -2\\ -1 & 2\end{bmatrix}$$

Sol:

$$AC=\begin{bmatrix}1 & 3\\ 0 & 1\end{bmatrix}\begin{bmatrix}1 & -2\\ -1 & 2\end{bmatrix}=\begin{bmatrix}-2 & 4\\ -1 & 2\end{bmatrix}$$

$$BC=\begin{bmatrix}2 & 4\\ 2 & 3\end{bmatrix}\begin{bmatrix}1 & -2\\ -1 & 2\end{bmatrix}=\begin{bmatrix}-2 & 4\\ -1 & 2\end{bmatrix}$$

So $AC=BC$

But $A\neq B$

2.3 The Inverse of a Matrix

- Inverse matrix:

Consider $A\in M_{n\times n}$

If there exists a matrix $B\in M_{n\times n}$ such that $AB=BA=I_n$,

then (1) A is **invertible** (or **nonsingular**)

(2) B is **the inverse** of A

- Notes:

(1)A matrix that does not have an inverse is called **noninvertible** (or **singular**).

(2) The inverse of A is unique and denoted by A^{-1}

(3) $AA^{-1}=A^{-1}A=I$

- Example: Find the inverse of the following matrix

$A = \begin{pmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{pmatrix}$. We have to find a matrix $\begin{pmatrix} x_{11} & x_{12} & x_{13} \\ x_{21} & x_{22} & x_{23} \\ x_{31} & x_{32} & x_{33} \end{pmatrix}$

satisfying $\begin{pmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{pmatrix} \begin{pmatrix} x_{11} & x_{12} & x_{13} \\ x_{21} & x_{22} & x_{23} \\ x_{31} & x_{32} & x_{33} \end{pmatrix} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$ (c_1, c_2, c_3; b_1, b_2, b_3)

That is, we have to solve three systems of linear equations:

$$\begin{pmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{pmatrix} \begin{pmatrix} x_{11} \\ x_{21} \\ x_{31} \end{pmatrix} = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} \qquad \begin{pmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{pmatrix} \begin{pmatrix} x_{12} \\ x_{22} \\ x_{32} \end{pmatrix} = \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}$$

$$\begin{pmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{pmatrix} \begin{pmatrix} x_{13} \\ x_{23} \\ x_{33} \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}$$

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Equivalently, in the matrix form, we have

$A.c_1 = b_1$,

$A.c_2 = b_2$,

$A.c_3 = b_3$.

Observation: Three system have the same coefficient matrix. So, if we use Gauss-Jordan elimination method, we can solve them simultaneously.

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- Find the inverse of a matrix by Gauss-Jordan Elimination:

$$[A \mid I] \xrightarrow{\text{Gauss-Jordan Elimination}} [I \mid A^{-1}]$$ I is identity matrix

- Ex 2: (Find the inverse of the matrix)

$$A = \begin{bmatrix} 1 & 4 \\ -1 & -3 \end{bmatrix}$$

Sol:

$$AX = I$$

$$\begin{bmatrix} 1 & 4 \\ -1 & -3 \end{bmatrix}\begin{bmatrix} x_{11} & x_{12} \\ x_{21} & x_{22} \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

$$\begin{bmatrix} x_{11}+4x_{21} & x_{12}+4x_{22} \\ -x_{11}-3x_{21} & -x_{12}-3x_{22} \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

$$\Rightarrow \begin{array}{rcrcl} x_{11} & + & 4x_{21} & = & 1 \\ -x_{11} & - & 3x_{21} & = & 0 \end{array} \quad (1)$$

$$\begin{array}{rcrcl} x_{12} & + & 4x_{22} & = & 0 \\ -x_{12} & - & 3x_{22} & = & 1 \end{array} \quad (2)$$

$$(1) \Rightarrow \left[\begin{array}{cc|c} 1 & 4 & 1 \\ -1 & -3 & 0 \end{array}\right] \xrightarrow{r_{12}^{(1)},\ r_{21}^{(-4)}} \left[\begin{array}{cc|c} 1 & 0 & -3 \\ 0 & 1 & 1 \end{array}\right] \Rightarrow x_{11} = -3, x_{21} = 1$$

$$(2) \Rightarrow \left[\begin{array}{cc|c} 1 & 4 & 0 \\ -1 & -3 & 1 \end{array}\right] \xrightarrow{r_{12}^{(1)},\ r_{21}^{(-4)}} \left[\begin{array}{cc|c} 1 & 0 & -4 \\ 0 & 1 & 1 \end{array}\right] \Rightarrow x_{12} = -4, x_{22} = 1$$

Thus

$$X = A^{-1} = \begin{bmatrix} x_{11} & x_{12} \\ x_{21} & x_{22} \end{bmatrix} = \begin{bmatrix} -3 & -4 \\ 1 & 1 \end{bmatrix} \quad (AX = I = AA^{-1})$$

- Note:

$$\left[\begin{array}{cc:cc} 1 & 4 & 1 & 0 \\ -1 & -3 & 0 & 1 \end{array}\right] \xrightarrow[r_{12}^{(1)}, r_{21}^{(-4)}]{\text{Gauss–JordanElimination}} \left[\begin{array}{cc:cc} 1 & 0 & -3 & -4 \\ 0 & 1 & 1 & 1 \end{array}\right]$$

$$A \qquad I \qquad\qquad I \qquad A^{-1}$$

If A can't be row reduced to I, then A is singular.

- Ex 3: (Find the inverse of the following matrix)

$$A = \begin{bmatrix} 1 & -1 & 0 \\ 1 & 0 & -1 \\ -6 & 2 & 3 \end{bmatrix}$$

Sol:

$$[A \vdots I] = \left[\begin{array}{ccc:ccc} 1 & -1 & 0 & 1 & 0 & 0 \\ 1 & 0 & -1 & 0 & 1 & 0 \\ -6 & 2 & 3 & 0 & 0 & 1 \end{array}\right]$$

$$\xrightarrow{r_{12}^{(-1)}} \left[\begin{array}{ccc:ccc} 1 & -1 & 0 & 1 & 0 & 0 \\ 0 & 1 & -1 & -1 & 1 & 0 \\ -6 & 2 & 3 & 0 & 0 & 1 \end{array}\right] \xrightarrow{r_{13}^{(6)}} \left[\begin{array}{ccc:ccc} 1 & -1 & 0 & 1 & 0 & 0 \\ 0 & 1 & -1 & -1 & 1 & 0 \\ 0 & -4 & 3 & 6 & 0 & 1 \end{array}\right]$$

$$\xrightarrow{r_{23}^{(4)}} \left[\begin{array}{ccc:ccc} 1 & -1 & 0 & 1 & 0 & 0 \\ 0 & 1 & -1 & -1 & 1 & 0 \\ 0 & 0 & -1 & 2 & 4 & 1 \end{array}\right] \xrightarrow{r_{3}^{(-1)}} \left[\begin{array}{ccc:ccc} 1 & -1 & 0 & 1 & 0 & 0 \\ 0 & 1 & -1 & -1 & 1 & 0 \\ 0 & 0 & 1 & -2 & -4 & -1 \end{array}\right]$$

$$\xrightarrow{r_{32}^{(1)}}\left[\begin{array}{ccc:ccc}1&-1&0&1&0&0\\0&1&0&-3&-3&-1\\0&0&1&-2&-4&-1\end{array}\right]\xrightarrow{r_{21}^{(1)}}\left[\begin{array}{ccc:ccc}1&0&0&-2&-3&-1\\0&1&0&-3&-3&-1\\0&0&1&-1&-4&-1\end{array}\right]$$

$= [\, I \,\vdots\, A^{-1} \,]$

So the matrix A is invertible, and its inverse is

$$A^{-1}=\begin{bmatrix}-2&-3&-1\\-3&-3&-1\\-2&-4&-1\end{bmatrix}$$

- Check:

$$AA^{-1}=A^{-1}A=I$$

- Power of a square matrix:

$$(1)\,A^0=I$$

$$(2)\,A^k=\underbrace{AA\cdots A}_{k\text{ factors}}\qquad (k>0)$$

$$(3)\,A^r\cdot A^s=A^{r+s}\qquad r,s:\text{integers}$$

$$(A^r)^s=A^{rs}$$

$$(4)\,D=\begin{bmatrix}d_1&0&\cdots&0\\0&d_2&\cdots&0\\\vdots&\vdots&\ddots&\vdots\\0&0&\cdots&d_n\end{bmatrix}\Rightarrow D^k=\begin{bmatrix}d_1^k&0&\cdots&0\\0&d_2^k&\cdots&0\\\vdots&\vdots&&\vdots\\0&0&\cdots&d_n^k\end{bmatrix}$$

Properties of inverse matrices:

If A is an invertible matrix, k is a positive integer, and c is a scalar not equal to zero, then

(1) A^{-1} is invertible and $(A^{-1})^{-1} = A$

(2) A^k is invertible and $(A^k)^{-1} = \underbrace{A^{-1}A^{-1}\cdots A^{-1}}_{k \text{ factors}} = (A^{-1})^k = A^{-k}$

(3) cA is invertible and $(cA)^{-1} = \frac{1}{c}A^{-1}, c \neq 0$

(4) A^T is invertible and $(A^T)^{-1} = (A^{-1})^T$

(5) If A and B are invertible matrices of size n, then AB is invertible nd $(AB)^{-1} = B^{-1}A^{-1}$

(6) If C is an invertible matrix, then

(a) If $AC=BC$, then $A=B$ (Right cancellation property)

(b) If $CA=CB$, then $A=B$ (Left cancellation property)

(7) Cramer's rule : If A is an invertible matrix, then the system of linear equations $Ax = b$ has a unique solution given by $x = A^{-1}b$

(7') If A is an $n \times n$ matrix, then the following statements are equivalent

(a) A is invertible.

(b) $A\mathbf{x} = \mathbf{b}$ has a unique solution for every $n\times1$ column matrix **b.**

(c) $A\mathbf{x} = \mathbf{0}$ has only the trivial solution.